

NAME

mbpy_xyplot – Create an executable shellscript which will generate a GMT modern-mode plot of xy data from one or more data files.

VERSION

Version 5.0

SYNOPSIS

```
mbpy_xyplot [-Ifile[:key=value...]] [-Ifile[:key=value...] ...] [-Oroot] [--formatfmt] [--titletext] [--xlabel-
text] [--ylabeltext] [--dpdpi] [-X] [-V]
[-Rxmin/xmax/ymin/ymax[r]] [-Jspec] [-Ppagesize] [--portrait | --landscape] [-Bspec] [--length-scale-
spec]
[-Ssymbol] [-Gcolor] [-Wpen]
[--coast]      [--coast-resolution{f,h,i,l,c}]      [--coast-dry-fillcolor]      [--coast-wet-fillcolor]
[--coast-lake-fillcolor] [--coast-rivercolor] [--coast-boundaryspec] [--coast-penpen]
[--textlon/lat/size/angle/font/justify/text]
[--gmt-defaultparam/value]
```

DESCRIPTION

mbpy_xyplot is a Python/PyGMT-era replacement for the Perl macro **mbm_xyplot**. It reads one or more xy data files and generates an executable shellscript of **GMT modern mode** commands (**gmt begin / gmt end**) which, when run, plots them as lines, symbols, or both, optionally overlaid with a coastline and/or text labels. Axes may be an auto-fit linear (Cartesian) scale or any GMT map projection.

The plot is scaled to fit the specified page size using the same page-layout, projection-fitting, and tick-interval heuristics as **mbm_histplot** -- confirmed by direct comparison of the two Perl macros' own sources to be identical (aside from whitespace), and distinct from **mbm_grdplot**'s own versions of the same computations (see **mbpy_histplot**(1)). In particular, when no **--projection** is given, **mbpy_xyplot** computes two *independent* x and y scales that each separately stretch to fill the page (unlike **mbpy_grdplot**/**mbpy_grd3dplot**'s single aspect-ratio-preserving scale, which would be wrong here since an xy plot's x and y axes routinely have unrelated units and ranges -- e.g. a time-series plot).

Unlike **mbm_xyplot**'s single-letter **-Msub:sub:...** grammar for coastline/text-label/length-scale tuning, **mbpy_xyplot** exposes each of those as its own descriptive long option (see OPTIONS below).

mbpy_xyplot never writes a persistent temporary file, unlike **mbm_xyplot** itself: for every input file, **mbm_xyplot** preprocesses (delimiter-splits and, if requested, applies a column-selection/math expression to) the data into a file named *<input><n>.<pid>* left behind in the current directory after every run, whether or not the plot script is ever executed. **mbpy_xyplot** instead performs the equivalent column-selection/math once in memory, in Python, to determine the auto-fit plot bounds, and at plot time pipes each input file through a small **awk**(1) program straight into **gmt plot**'s standard input -- a live, ephemeral pipe, with nothing written to disk.

Each **--input** file may carry its own colon-separated overrides, in *key=value* form: **symbol=**, **fill=**, **pen=** (styling, defaulting from **-S/-G/-W**), **delimiter=** (default: whitespace), and **xexpr=/yexpr=** (column-selection/math expressions using **mbm_xyplot**'s own **c[N]** field-selection and **#** line-number mini-language, e.g. **c[3]** or **c[3]*2+1**; default **xexpr=c[1]**, **yexpr=c[2]**). A line beginning with **>** is always passed through unchanged as a multi-segment marker, matching **mbm_xyplot**'s (and **gmt plot**'s) own convention.

A genuine bug was found and fixed while validating this port against real-world data: **mbm_xyplot**'s own per-line numeric-result check is unintentionally a no-op (its validation regex is satisfied by the empty string), so an ordinary text header line at the top of a data file passes through into the plotted file unfiltered -- and because

gmt plot infers each column's data type from the *first* record it reads, a single such non-numeric first line makes it silently discard every subsequent, perfectly valid row instead of just that one line. **mbpy_xyplot** validates each computed x/y pair with a real numeric check instead, skipping (with a warning) only the genuinely bad lines.

A related, deliberate addition with no equivalent in **mbm_xyplot** at all: a data column that is exactly constant (e.g. a real "heave" trace logged as 0.0 for an entire survey) collapses to a zero-width axis that neither classic nor modern GMT can project. **mbpy_xyplot** pads such a degenerate axis by a small margin so it still plots as a visible flat line instead of aborting.

mbpy_xyplot requires **GMT** 6 or later (modern mode), a POSIX **awk**(1), and a Python 3 interpreter. It depends on **mbpy_layout.py** and **baseline_gmt_defaults()** from **mbpy_grdplot_script.py** (shared with **mbpy_grdplot/mbpy_grd3dplot**), and reuses the page-size/tick-interval helper functions from **mbpy_histplot_script.py**, plus its own **mbpy_xyplot_script.py**, all installed alongside it.

MB-SYSTEM AUTHORSHIP

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OPTIONS

- I, --input** *file[:key=value...]*
 An xy data file to plot, with optional colon-separated overrides **symbol=**, **fill=**, **pen=**, **delimiter=**, **xexpr=**, **yexpr=** (see DESCRIPTION). Repeatable; required at least once.
- O, --output** *root*
 Output root name. Default: the first **--input** file's basename with its extension removed.
- format** *fmt*
 Output format passed to **gmt figure**: **pdf**, **png**, **ps**, **jpg**, **tif**, **eps**, ... Default **pdf**.
- title** *text*
 Plot title, placed above the frame.
- xlabel** *text*, **--ylabel** *text*
 Axis labels. Only used for a linear/Cartesian projection (an auto-fit default, or an explicit **-Jx/-JX**); a geographic or other map projection has no per-axis label, matching **mbm_xyplot**.
- dpi** *dpi*
 The final raster's resolution (via **gmt figure**).
- X, --execute**
 Run the generated script immediately after writing it.
- V, --verbose**
 Print the generated script's path and a one-line summary.
- R, --region** *xmin/xmax/ymin/ymax[r]*
 Plot region. Fields may be given as *dd:mm:ss*. A trailing **r** means the four values are two corner points (southwest, northeast) rather than a w/e/s/n range; a bare **r** reformats the auto-computed data bounds into that corner-point form. Default: computed from the data itself.

- J, --projection *spec***
GMT map scale/projection spec, e.g. **x0.02** (linear, 0.02 inch/unit), **m1.0** (Mercator, 1 inch/degree). Default: an auto-fit linear projection with independent x and y scales (see DESCRIPTION), chosen to fill the page.
- P, --pagesize *size***
Page size code: **a-f**, **e1** (US sizes), **a0-a10**, **b0-b10**, **c0-c7** (ISO/architectural), or **m1-m6** (large-format). Default **a**.
- portrait, --landscape**
Force the page orientation instead of choosing automatically; when neither is given, this auto-fit independent-scale mode defaults to landscape, matching **mbm_xyplot/mbm_histplot**.
- B, --tick-info *spec***
Raw **-B** basemap tick-interval spec, overriding the automatically computed one.
- length-scale *spec***
A GMT distance (map) scale bar spec, passed as **-Lspec** on **gmt basemap**, e.g. **fx10/2/32.70/5.0+lkm** (**mbm_xyplot**'s **-MGL**). The literal word **center** appearing between two slashes is replaced with the plot's own data center (e.g. **fxcenter/5.0+lkm**), matching **mbm_xyplot**'s own substitution.
- S, --symbol *symbol***
Default GMT symbol code and size for every **--input** file that does not set its own **symbol=**, e.g. **c0.1** (circle, 0.1 inch). Default: none (points are connected with a line instead).
- G, --fill *color***
Default symbol/polygon fill color for every **--input** file that does not set its own **fill=**. Default: none.
- W, --pen *pen***
Default line pen for every **--input** file that does not set its own **pen=**, e.g. **1p,black**. Default: none.
- coast**
Enable a coastline overlay (**gmt coast**), drawn *after* the xy data plots and *before* text labels and the basemap frame, matching **mbm_xyplot**'s own layer order exactly (data points that fall on a solid land fill are covered by it, just as in the classic macro).
- coast-resolution *{f,h,i,l,c}***
Coastline database resolution: full, high, intermediate, low, crude. Default **f** when **--coast** is enabled with no other coastline option given.
- coast-dry-fill, --coast-wet-fill, --coast-lake-fill *color***
Fill colors for land, water, and lakes. **--coast-dry-fill** defaults to **200** (a mid-gray) when **--coast** is enabled with no fill, pen, river, or boundary option given at all -- an **mbm_xyplot**-specific default with no equivalent in **mbm_grdplot/mbpy_grdplot**.
- coast-river *color***
Draw rivers in the given color.
- coast-boundary *spec***
Draw a national boundary; repeatable.
- coast-pen *pen***
Pen for the coastline itself. Defaults to **1p** together with the **--coast-dry-fill** default described above.
- text *lon/lat/size/angle/font/justify/text***
Place a text label (**gmt text**), e.g. **-122.2/36.6/12/0/1/LB/Monterey Bay**. Repeatable.
- gmt-default *param/value***
Override a GMT default parameter for this plot (e.g. **MAP_FRAME_TYPE/plain**); repeatable.

EXAMPLES

Plot a single time series of sensor depth values as a thin connecting line, rendered directly to a PNG:

```
mbpy_xyplot -I "depth_vs_time.txt:pen=0.5p,blue" -O depth_ts \
  --title "Sensor Depth vs Time" --xlabel "Time (s)" \
  --ylabel "Sensor Depth (m)" \
  --pagesize b --landscape --format png --dpi 150 --execute -V
```

Plot navigation fixes from a comma-delimited file (selecting the longitude and latitude columns explicitly) as symbols, with a coastline underlay:

```
mbpy_xyplot -I "fixes.csv:delimiter=,:xexpr=c[2];yexpr=c[3]:symbol=c0.05:fill=red" \
  -O fixes_plot --title "Vehicle Fixes" \
  --coast --coast-resolution i \
  --format png --execute -V
```

Plot two data files together, one as a line and one as symbols, over a user-specified region:

```
mbpy_xyplot -I "reference.xy:pen=1p,black" \
  -I "observed.xy:symbol=c0.08:fill=red" \
  -O comparison -R "-10/110/-5/55" \
  --title "Reference vs Observed" --format pdf -V
```

SEE ALSO

mbm_xyplot(1), **mbpy_grdplot(1)**, **mbpy_grd3dplot(1)**, **mbpy_histplot(1)**, **awk(1)**, **mbdefaults(1)**, **mb-system(1)**

BUGS

Please report any bugs to the MB-System project on GitHub (<https://github.com/dwcaress/MB-System>).